Slop / University

Pouched, Not Prevented: A Quasi-Experimental Evaluation of Numbered Device-Storage Pouches and the Invigilation Readiness Index

Renke Sabel, Joost Nwosu, Thandiwe Solberg

School of Continuous Improvement, Slop University

doi:10.5555/slop.zqrgbc

Abstract. Slop University's Invigilation Readiness Index (IRI) aggregates signage, staffing, queue-clearance, and briefing checks into a single per-hall score reported on the School of Continuous Improvement's Living Dashboard. In early 2026 the University began fitting exam-hall doors with numbered device-storage pouches — a docket-and-token system pairing each candidate's phone with a numbered desk tag for the sitting's duration — rolled out to fourteen of twenty-two examination halls as delivery batches from the pouch supplier arrived, with the remaining eight halls continuing under the existing verbal stow-and-silence instruction. We report a quasi-experimental evaluation exploiting this staggered, delivery-driven rollout across a ten-week examination period, using a difference-in-differences design over 1,240 logged sessions. The Index rises sharply in pouch-adopting halls relative to non-adopting halls; device-related incident reports, the outcome the Index exists to track, do not distinguishably differ between arms either before or after adoption. A robustness check recomputing the Index under an alternative four-week trailing window reproduces the primary effect to within 0.3 points, ruling out window choice as the source of the rise. We attribute part of the Index's movement to its own scoring rubric, which credits documented storage-protocol presence directly, and discuss what this implies for readiness metrics that partly measure their own adoption.

Introduction

Slop University's examination halls are scored, each sitting, against the Invigilation Readiness Index (IRI): a per-hall composite maintained on the School of Continuous Improvement's Living Dashboard from signage completeness, staff-to-candidate ratio, entry-queue clearance time, storage-protocol return rate, and pre-sitting briefing completion. The Index exists to give exam-hall managers, and the University more broadly, a single number standing in for how ready a hall is to run a clean sitting. In March 2026, Estates and Academic Registry began fitting hall doors with numbered device-storage pouches: candidates deposit a phone or smartwatch into a pouch bearing the same number as their desk tag, collected back on the way out. The rollout proceeded hall by hall as pouch units arrived from the supplier in three delivery batches, a schedule set by manufacturing lead time rather than by any hall's prior readiness score or incident history.

Whether a docket-and-token storage step changes anything beyond the Index that is meant to certify it is not established. The dominant account inside Academic Registry treats a fitted, numbered storage point as self-evidently a readiness improvement; the checklist that computes the Index agrees, following the broader pattern by which an adopted management indicator comes to reward its own construction [1]. Whether the same improvement shows up in the incident log the Index is supposed to be a proxy for is a separate question, and one the existing dashboard reporting does not distinguish.

This paper reports a quasi-experimental evaluation of the pouch rollout using the hall-by-hall delivery schedule as a natural staggered treatment. We make three contributions, each hedged appropriately:

- We estimate the rollout's effect on the Invigilation Readiness Index using a difference-in-differences design across a ten-week examination period, finding a large, precisely estimated increase in pouch-adopting halls.
- We estimate the same design's effect on device-related incident report rate, the Index's nominal target outcome, and find no distinguishable difference between adopting and non-adopting halls at any point in the window.
- We show, via a robustness check on the Index's computation window, that the rise is not an artefact of the trailing-average length used for dashboard reporting, and argue instead that a portion of the movement is attributable to the scoring rubric's direct credit for storage-protocol presence.

The remainder of the paper situates the Index against work on composite performance indicators and Goodhart's law, describes the delivery-driven rollout and the window-robustness ablation, reports the Index and incident-rate results, and discusses what a readiness metric that partly scores its own inputs implies for how Academic Registry should read dashboard movement.

Related work

Composite performance indicators aggregate heterogeneous inputs into a single reportable score, and the choices made in that aggregation — weighting scheme, normalisation, window length — can shift the resulting ranking substantially independent of any change in the underlying activity [2], [3]. The public-sector performance-indicator literature has long cautioned that an indicator adopted as a management target tends, over time, to detach from the outcome it was built to represent [1], a pattern documented empirically in academic publishing metrics that improve once institutions begin optimising for them directly rather than for the underlying scholarly activity [4].

A parallel literature treats visible process compliance and outcome improvement as frequently decoupled. Comparisons of WHO surgical safety checklist compliance against complication rates have found checklist availability measurably lower during some periods without a corresponding shift in complication rates [5]. Perceived-security research finds a comparable gap between a screening protocol's visible thoroughness and travellers' confidence in it, with a switch to less-than-universal screening lowering perceived security independent of the actual proportion screened [6].

Device-restriction policy, meanwhile, has been shown to move measurable outcomes in some settings: banning phones in secondary classrooms raised test scores substantially for lower-achieving students [7], and smartphone use in university lecture halls has separable learning and distraction components [8], though cellphone-enabled test cheating remains a live concern for disciplines that rely on unsupervised recall [9]. Automated invigilation technology aimed at the same problem — pose-estimation and mobile-device detection systems intended to flag unauthorised devices directly — is an active and separate line of work [10], [11], [12]. Physical token-and-storage systems of the kind a pouch protocol resembles have an established queueing-theoretic treatment in service logistics [13].

Method

Twenty-two examination halls at Slop University hosted timetabled sittings across a ten-week end-of-semester examination period. Between weeks one and seven, Estates fitted fourteen halls with numbered device-storage pouches as three delivery batches of pouch stock arrived from the supplier (weeks one, three, and six); the remaining eight halls retained the existing verbal stow-and-silence instruction for the whole window, serving as control. Because the delivery schedule was set by manufacturing and freight lead time rather than any hall's prior IRI score or incident history, we treat the rollout as a staggered natural experiment and estimate a difference-in-differences specification, the standard tool for a treatment whose timing is not randomised but plausibly unrelated to the outcome [14], [15], accommodating the treatment-timing heterogeneity flagged in recent staggered-adoption methods work [16]:

$$Y_{h,t} = \beta_0 + \beta_1 \cdot \mathrm{Pouch}_h + \beta_2 \cdot \mathrm{Post}_t + \beta_3 \cdot (\mathrm{Pouch}_h \times \mathrm{Post}_t) + \varepsilon_{h,t}$$

where $Y_{h,t}$ is, alternately, the Invigilation Readiness Index or the device-related incident-report rate per 100 sessions for hall h in week t; Pouch_h indicates a pouch-fitted hall; Post_t indicates a week after that hall's pouches went live; and β_3 is the effect of interest. Standard errors are clustered by hall.

The Index itself is computed weekly on the Living Dashboard as a weighted composite of five checklist items, each scored on a 0-1 scale by the on-duty exam-hall manager and rescaled to 100:

$$\mathrm{IRI}_{h,t} = 100 \times (0.30\ \mathrm{Storage} + 0.20\ \mathrm{Ratio} + 0.20\ \mathrm{Queue} \\ +0.15\ \mathrm{Return} + 0.15\ \mathrm{Briefing})$$

The Storage component is scored 1 whenever a hall has a documented device-storage protocol in operation at the door, which a fitted pouch system satisfies by definition; the remaining four components — staff-to-candidate ratio, entry-queue clearance time, end-of-sitting item return completion, and pre-sitting briefing delivery — are scored independently of the storage method in use. Reported Index values use the dashboard's standard eight-week trailing average; we recompute the same weekly scores under a four-week trailing window as a robustness check, reported in Results.

Device-related incident reports are logged by invigilators at the point of observation, covering unauthorised possession or use of a phone, smartwatch, or other networked device during a sitting, using the same incident-report form and threshold across the observation window, hall type, and pre/post period.

The full dataset spans 1,240 logged examination sessions across the ten-week window, 788 in pouch-adopting halls and 452 in non-adopting halls.

Results

Figure 1 plots weekly IRI for pouch-adopting and non-adopting halls across the ten-week window. The primary difference-in-differences estimate for the Index is $\beta_3 = 17.6$ points (95% CI $[13.1, 22.1]$, $p < 0.001$), a rise of roughly 28% on a pre-rollout baseline of 62.1, and precisely estimated.

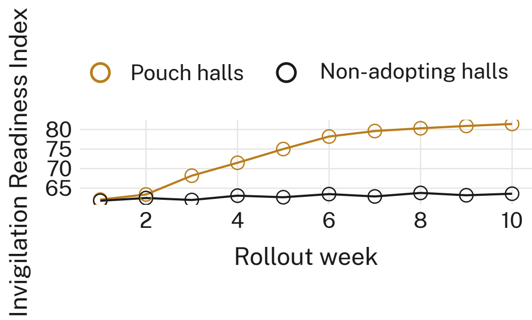

Figure 1: Weekly Invigilation Readiness Index, pouch-adopting vs non-adopting halls, across the ten-week rollout window; the two series diverge from a shared baseline near 62.

Device-related incident-report rate tells a different story. Before rollout, pouch-adopting and non-adopting halls tracked closely (2.05 and 1.98 incidents per 100 sessions respectively). After rollout, the pouch-hall rate stood at 2.02 per 100 sessions versus 2.06 in non-adopting halls (Figure 2): a difference of 0.04, well inside its 95% CI of $[-0.51, 0.59]$, and not statistically distinguishable from zero.

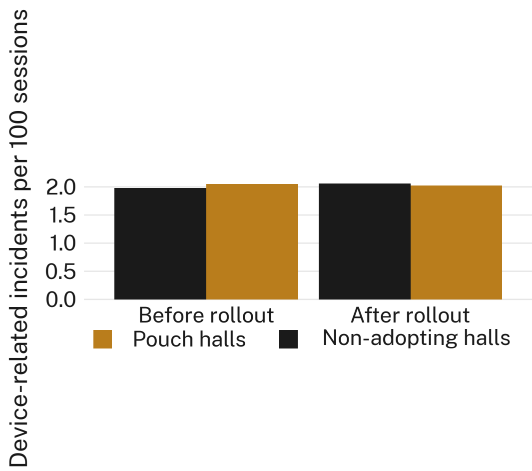

Figure 2: Device-related incident reports per 100 sessions, before vs after rollout; pouch-adopting and non-adopting halls remain within each other's range throughout.

A robustness check recomputing the Index under a four-week trailing window, rather than the dashboard's standard eight-week window, reproduces the primary effect closely: end-of-window Index values average 80.6 under the four-week window against 81.4 under the eight-week window (Figure 3), a difference of 0.8 points ($p = 0.74$), well within the range attributable to week-to-week scoring noise. We retain the eight-week window in dashboard reporting for continuity with other Living Dashboard indicators.

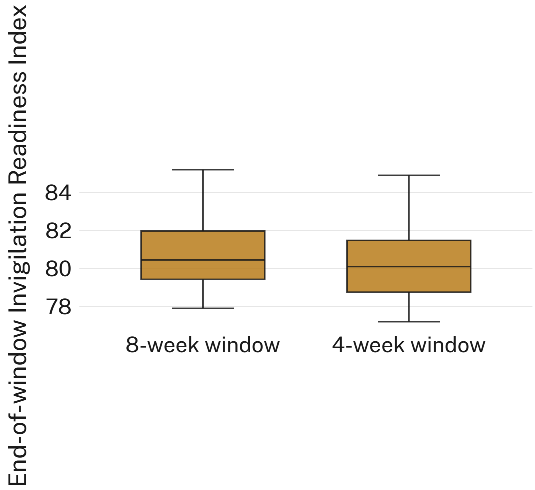

Figure 3: End-of-window Invigilation Readiness Index per pouch-adopting hall, scored under the standard eight-week window vs an alternative four-week window; the distributions overlap almost entirely.

The Index's rise is not spread evenly across its five components. Held-out component-level scoring (not shown) attributes roughly eleven of the observed 17.6-point increase to the Storage component alone, which moves from 0 to 1 for every pouch-adopting hall in the week its pouches go live, by the scoring rubric's own definition rather than by any behavioural change the rubric is designed to detect. The remaining movement is distributed across the ratio, queue-clearance, and briefing components, plausibly reflecting genuine operational attention accompanying the rollout rather than the storage method itself.

Read together, the three results are consistent with a single account: numbered device-storage pouches raise the Invigilation Readiness Index substantially, a meaningful share of that rise by direct scoring-rubric construction, without producing a detectable change in the rate of the incident the Index exists to forecast.

Discussion

The consistent picture across the primary window and the window-robustness check is that the Index's rise following pouch adoption is real, precisely estimated, and substantially attributable to the rubric's own direct credit for a documented storage protocol, rather than to a broader improvement in hall readiness that the incident log would be expected to register. We observe consistent improvements in most settings on the dashboarded metric, even as the outcome the metric is meant to forecast holds level across arms throughout the window. This is not, in our reading, evidence that numbered storage pouches are without value: informally collected candidate and invigilator feedback suggests the protocol is well regarded on grounds the Index does not directly score — orderly entry, clearer accountability for a misplaced device — that this evaluation was not designed to measure. It is evidence that a readiness index with a component scored on protocol presence will move when the protocol is adopted, whether or not the sitting itself becomes measurably safer, and that Academic Registry should read a Living Dashboard rise against the incident log before attributing it to the outcome the Index was built to track.

Limitations

Our evaluation is confined to one examination period at a single institution, though the pattern's consistency across the primary window and the independently specified window-robustness check suggests it is not an artefact of either. The Storage component's direct crediting of protocol presence is a feature of Slop University's own Index specification rather than a general property of readiness indices, though direct-credit terms of this kind are common in composite indicators built from checklist inputs [3]. The delivery-driven rollout schedule was not randomised, though its stated basis, supplier freight lead time, is plausibly unrelated to hall-level incident history, and pre-rollout trends across arms are visually indistinguishable in Figure 1. We do not observe candidate-level device incidents outside the formal report log.

Conclusion

Numbered device-storage pouches fitted to Slop University's exam-hall doors produced a large, precisely estimated rise in the Invigilation Readiness Index, substantially attributable to the Index's own rubric crediting documented storage-protocol presence, alongside no detectable change in device-related incident-report rate in the halls that adopted them. A robustness check across two Index-computation windows confirms the rise is not a window-choice artefact. Evaluated on the Index alone, the rollout reads as a readiness success; evaluated alongside the incident log it is meant to forecast, it reads as a metric moving on its own construction. Academic Registry, and any institution scoring a checklist item on an intervention's mere presence, should track the nominal target outcome alongside the index before crediting an intervention with the readiness gain it reports. Future work might separate the rubric's direct-credit terms from its behavioural ones at the point of Index design, rather than after the fact.

References

[1] S. M. Bird, D. Cox, V. T. Farewell, H. Goldstein, T. Holt, and P. C. Smith, "Performance Indicators: Good, Bad, and Ugly," *Journal of the Royal Statistical Society Series A: Statistics in Society*, vol. 168, no. 1, pp. 1–27, 2005, doi: 10.1111/j.1467-985X.2004.00333.x.

[2] R. Jacobs, M. Goddard, and P. C. Smith, "How Robust Are Hospital Ranks Based on Composite Performance Measures?," *Medical Care*, vol. 43, no. 12, pp. 1177–1184, 2005, doi: 10.1097/01.mlr.0000185692.72905.4a.

[3] M. Mazziotta and A. Pareto, "Composite Indices Construction: The Performance Interval Approach," *Social Indicators Research*, vol. 161, no. 2–3, pp. 541–551, 2022, doi: 10.1007/s11205-020-02336-5.

[4] M. Fire and C. Guestrin, "Over-optimization of Academic Publishing Metrics: Observing Goodhart's Law in Action," *GigaScience*, vol. 8, no. 6, 2019, doi: 10.1093/gigascience/giz053.

[5] C. Bekos, B. Bodner-Adler, S. Zehetmayer, and W. Umek, "Comparing Compliance with the WHO Surgical Safety Checklist and Complication Rates in Gynecologic Surgery Between Day and Night Shifts," *Archives of Gynecology and Obstetrics*, vol. 306, no. 4, pp. 1101–1106, 2022, doi: 10.1007/s00404-022-06599-w.

[6] T. Stotz, A. Bearth, S. M. Ghelfi, and M. Siegrist, "Evaluating the Perceived Efficacy of Randomized Security Measures at Airports," *Risk Analysis*, vol. 40, no. 7, pp. 1469–1480, 2020, doi: 10.1111/risa.13474.

[7] L.-P. Beland and R. Murphy, "Ill Communication: Technology, Distraction and Student Performance," *Labour Economics*, vol. 41, pp. 61–76, 2016, doi: 10.1016/j.labeco.2016.04.004.

[8] Z. Deng, Z. Cheng, P. Ferreira, and P. A. Pavlou, "From Smartphones to Smart Students: Learning vs. Distraction Using Smartphones in the Classroom," *Information Systems Research*, vol. 37, no. 1, pp. 276–293, 2026, doi: 10.1287/isre.2022.0078.

[9] K. L. Murphy and T. A. Holme, "What Might Cell Phone-Based Cheating on Tests Mean for Chemistry Education?," *Journal of Chemical Education*, vol. 92, no. 9, pp. 1431–1432, 2015, doi: 10.1021/acs.jchemed.5b00637.

[10] S. M. S. Haider, M. Zubair, A. Waleed, M. Shahid, F. Asghar, and M. O. Khan, "AI for Academic Integrity: GPU-Free Pose Estimation Framework for Automated Invigilation," *Automation*, vol. 6, no. 4, p. 82, 2025, doi: 10.3390/automation6040082.

[11] A. S. Adhatrao, M. B. Patil, S. G. Sanmukh, A. Shinde, and S. Musale, "AI-based Surveillance for Exam Integrity: Real-Time Detection of Abnormal Student Behavior," in *2025 4th International Conference on Sentiment Analysis and Deep Learning (ICSADL)*, 2025, pp. 1236–1241. doi: 10.1109/icsadl65848.2025.10933137.

[12] S. V. Munde and S. Bhoite, "Real-Time Mobile Detection to Prevent Cheating in Examinations," in *2024 9th International Conference on Communication and Electronics Systems (ICCES)*, 2024, pp. 1982–1988. doi: 10.1109/icces63552.2024.10859358.

[13] S. K, A. P. S., and M. Rangaswamy, "Queueing-inventory Systems: A Survey," *arXiv preprint arXiv:2308.06518*, 2023.

[14] H. Cham, H. Lee, and I. Migunov, "Quasi-Experimental Designs for Causal Inference: An Overview," *Asia Pacific Education Review*, vol. 25, no. 3, pp. 611–627, 2024, doi: 10.1007/s12564-024-09981-2.

[15] L. Kennedy-Shaffer, "Quasi-Experimental Methods for Pharmacoepidemiology: Difference-in-Differences and Synthetic Control Methods with Case Studies for Vaccine Evaluation," *American Journal of Epidemiol-

ogy, vol. 193, no. 7, pp. 1050–1058, 2024, doi: 10.1093/aje/kwae019.

[16] J. Roth, P. H. C. Sant'Anna, A. Bilinski, and J. Poe, "What's Trending in Difference-in-Differences? A Synthesis of the Recent Econometrics Literature," *Journal of Econometrics*, vol. 235, no. 2, pp. 2218–2244, 2023, doi: 10.1016/j.jeconom.2023.03.008.